Advanced Candidate Ranking Engine

A Production-Grade Hybrid Retrieval & Re-ranking Pipeline

Designed for the 'Senior AI Engineer - Founding Team' Role | 100k Candidates Pool

SUBMITTED BY:

Team Name: team_2+

Primary Contact: Anmol Raj (anmolraj499@gmail.com)

SYSTEM COMPLIANCE:

[X] 100% Offline (Zero API Calls at Runtime)

[X] CPU-Only execution (<= 16GB RAM constraint)

[X] Execution time: 193.75s (under 5-minute limit)

EXECUTIVE SUMMARY

Solving the Limitations of Traditional ATS

THE CRITICAL FLAWS OF KEYWORD FILTERS

- **1. Vulnerability to Keyword Stuffing:**
  Candidates with weak skills stuff their resumes with target keywords, pushing low-quality profiles to the top.
- **2. Semantic Blindness:**
  Fails to recognize conceptual synonyms. A candidate writing 'neural representation learning' is filtered out if the system only looks for 'embeddings'.
- **3. Ignoring Career Context:**
  Treats skills list in isolation, failing to evaluate career trajectories, tenure stability, and notice period constraints.
- **4. Honeypot Ingestion:**
  Standard databases are flooded with fake timelines, invalid experience declarations, and resume fraud (honeypot profiles).

THE TEAM_2+ PHILOSOPHY

- **1. Hybrid Search Architecture:**
  Combine fast keyword index funnels with deep contextual sentence-transformers to capture implicit expertise.
- **2. Hard Data Integrity Gates:**
  Verify candidate consistency by cross-referencing graduation years, skill durations, and job title coherence before scoring.
- **3. Behavioral Signal Fusion:**
  Fuses Redrob signals (notice periods, recruiter responsiveness, and geographic flexibility) directly into the mathematical rank.
- **4. Factual Explainability:**
  Generate honest, non-hallucinated explanations summarizing why a candidate is ranked, directly highlighting experience metrics.

SYSTEM ARCHITECTURE

The Decoupled Multi-Stage Funnel Design

PIPELINE WORKFLOW & COMPUTATION STAGES

1	**Stream Ingestion**	Ingests 100k records line-by-line to prevent memory leaks. Screens timeline/skills fraud.
2	**BM25 Lexical Funnel**	Computes fast keyword score over target stack (PyTorch, RAG). Narrows pool to top 2,000.
3	**Dense Embedding**	Computes cosine similarity with local SBERT model (all-MiniLM-L6-v2) on CPU.
4	**Signal Fusion**	Applies multipliers for notice period agility, relocation, and recruiter activity.
5	**Cross-Encoder**	Computes deep joint context overlap for top 200 using local ms-marco-MiniLM-L-6-v2.
6	**Shortlist Export**	Applies deterministic tie-breakers and outputs format-compliant CSV.

DATA QUALITY GUARDRAILS

Identifying Timeline Fraud and Honeypots

HONEYPOT DETECTION ALGORITHMS

- **Rule 1: Graduation/Experience Mismatch**
  Excludes profiles claiming total years of experience exceeding the span since graduation by more than 2 years:
  Exp > (2026 - First Graduation Year) + 2

- **Rule 2: Skill Duration Inflation**
  Excludes candidates claiming 'expert' or 'advanced' proficiency in 4 or more distinct skills but declaring exactly 0 months of duration.

- **Rule 3: Career Timeline Coherence**
  Cross-references job start and end dates with declared job durations, checking for temporal mismatches (>12 months discrepancy).

- **Rule 4: Semantic Drift Integrity Gate**
  Detects copy-paste profile descriptions (e.g. 'marketing manager' summaries paired with unrelated developer or mechanical engineering titles).

FILTERING IMPACT

35,208

Honeypot Candidates Screened Out

From the original 100,000 profile pool, 35,208 candidates were flagged as having inconsistent timeline claims or fraudulent skills sheets.

0%

Honeypots in Final Shortlist

Our strict programmatic logic guarantees that none of these anomalous profiles make it into the final recommended Top 100 shortlist.

THE TWO-STAGE RETRIEVAL MODEL

Scaling Semantic Comprehension on CPU

STAGE 1: BM25 LEXICAL RETRIEVAL

- **Mechanism:**
  Computes a document index over normalized terms in candidate summaries, titles, and career descriptions.
- **Query Vector:**
  Curated keyword list containing 'pytorch', 'vector database', 'faiss', 'qdrant', 'RAG', 'transformers', and 'mlops'.
- **Speed & Scale:**
  Filters candidate pool from 64,792 active profiles down to the top 2,000 in just 10.01 seconds.
- **Role:**
  Acts as a broad recall filter to ensure high-coverage of primary technical stack requirements.

STAGE 2: LOCAL SBERT EMBEDDINGS

- **Model selection:**
  Uses local 'all-MiniLM-L6-v2' model to map candidate profiles into a 384-dimensional vector space.
- **Zero-Network Constraint:**
  Loads model files locally from checkpoints inside the project. Requires zero network/API requests.
- **Computation:**
  Computes exact cosine similarity on CPU across the top 2,000 candidates in 15 seconds.
- **Impact:**
  Identifies matching candidates based on semantic meaning rather than exact word matches, preventing keyword stuffing.

FINAL OPTIMIZATION & RE-RANKING

Fusing Platform Signals with Joint Attention

STAGE 3: BEHAVIORAL SIGNAL MODULATION

- **1. Seniority Fit Curve:**
  Underqualified candidates receive steep penalty curves; overqualified profiles decay slowly to prevent screening out.
- **2. Notice Period Agility:**
  Bonus (+5%) for sub-15 days notice, penalty (-15%) for 90-day locks to account for drop-off hazard.
- **3. Location Fit Multipliers:**
  1.0 for target Indian hubs (Noida, Pune, NCR). 0.95 for relocation-willing. Penalty for non-willing remote candidates.
- **4. Platform Completeness:**
  Integrates recruiter response rates and profile completeness directly into the scoring loop.

STAGE 4: LOCAL CROSS-ENCODER RERANK

- **Dual-Encoder Correction:**
  Uses 'ms-marco-MiniLM-L-6-v2' locally on the top 200 candidates to refine the final shortlist.
- **Mechanism:**
  Computes joint cross-attention across candidate summaries and JD requirements simultaneously, extracting complex matching patterns.
- **Deterministic Tie-Breaking:**
  Uses alphabetical candidate ID sort as a tie-breaker when final scores are identical, ensuring stable and deterministic ranks.
- **Factual Rationale Generator:**
  Explains the specific title, years of experience, top matching tools, and notice period constraints without hallucinated claims.

Project Delivery & Verification Summary

SUBMISSION BENCHMARKS

- Execution Time: 193.75 seconds to screen, rank, and format 100,000 candidates on a single core.

- Memory Footprint: Peak RAM is well under 16GB due to line-by-line streaming of candidates.

- Formatting Verification: Passed all validate_submission.py checks, guaranteeing monotonic non-increasing scores, strict ranks (1-100), and alphabetical tie-breaking.

- Zero API Dependency: Operates completely offline, protecting candidate privacy and securing sandbox replicability.

DELIVERABLES CHECKLIST

[X] GitHub Repository (clean history, fully documented README)

[X] final shortlist CSV (team_2+.csv, validated format)

[X] Submission Metadata (submission_metadata.yaml)

[X] Google Colab Reproducibility Sandbox Notebook

[X] Slide Deck PDF (Methodology_team_2+.pdf)

Thank you! The pipeline is ready for submission to the Redrob Portal.